Claude and ChatGPT in Positron

International Marketing Analytics

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Today: Claude and ChatGPT in Positron

Pick a route → Posit Assistant → Extension → Terminal → MCP → Use it safely

Claude

- Anthropic's models
- Claude Code: the coding agent
- Pro from 20 USD a month, or an API key

ChatGPT

- OpenAI's models
- Codex: the coding agent
- ChatGPT account, or an API key

Course rules

- Optional: Copilot stays the default
- Nobody has to pay
- Same rules for every AI

Optional: GitHub Copilot in Posit Assistant remains the course default and is free with GitHub Education.

Four routes at a glance

Route	What you get	Claude needs	ChatGPT needs
1 Posit Assistant	chat in Positron's own AI pane	API key	API key
2 Vendor extension	an agent panel in Positron	Pro or Max	ChatGPT account
3 Terminal	the agent in Positron's terminal	Pro or Max	ChatGPT account
4 MCP (experimental)	agent works in your live session	with route 2 or 3	with route 2 or 3



No code completions: the grey suggestions while typing come only from GitHub Copilot or Posit AI Pass, not from Claude or ChatGPT.

Subscription or API key: the key question

Subscription (monthly)

- Claude Pro or Max, ChatGPT Plus or Pro
- Works in: extension and terminal (routes 2, 3)
- Not in Posit Assistant
- Flat price, usage limits

API key (pay per use)

- From platform.claude.com or platform.openai.com
- Works in: Posit Assistant, also routes 2 and 3
- Prepaid credit, billed per token
- Separate from any subscription

Common surprise: “I pay for Claude Pro, why does Positron want a key?” Because Pro does not include API access.

Route 1: Posit Assistant with an API key

Part 1 of 5 · get a key, connect it, what it costs

Get an Anthropic (Claude) API key

Claude Console

1. Open platform.claude.com
2. **Settings → Billing:** buy a small credit, e.g. 5 USD
3. **Set a spend limit;** auto-reload off
4. **Settings → API keys:** Create Key, then Add
5. **Copy the key** into a password manager

Good to know

- Credits expire after one year
- Claude Pro does not include API credit
- API use is not used for training

The key is a password: never in code, a .qmd file, GitHub or a chat.

Get an OpenAI (ChatGPT) API key

OpenAI platform

1. Open platform.openai.com
2. **Billing:** add prepaid credit, at least 5 USD
3. **Switch off** auto-recharge; set a budget
4. **API keys:** Create new secret key
5. **Copy the key** into a password manager

Good to know

- ChatGPT Plus does not include API credit
- Any OpenAI-compatible service also works
- API use is not used for training by default

Leaked a key? delete it on the platform at once and create a new one.

Connect the key in Positron

In Positron

1. **Command Palette:** Ctrl/Cmd + Shift + P
2. **Run:** Authentication: Configure Language Model Providers
3. **Pick** Anthropic or OpenAI
4. **Paste** the API key; it shows as connected
5. **View: Show Posit Assistant:** pick a model

Command Palette

```
Authentication:
  Configure Language
  Model Providers
```

Stored safely: in the system keychain, not in a file

Models: whatever your key may use

Same pane as Copilot: everything in the Copilot guide on Posit Assistant applies.

What a question costs

Claude model	Input (per 1M tokens)	Output (per 1M tokens)	One question*
Haiku 5.5	0.10 USD	0.50 USD	≈ 0.1 cent
Sonnet 5.5	2 USD	10 USD	≈ 2 cents
Opus 5.5	4 USD	20 USD	≈ 4 cents

* 5,000 tokens in, 1,000 out; own calculation from Anthropic's prices.

Chat is cheap, agents are not: an agent that reads your whole project can cost dollars

Protect yourself: small prepaid credit, a spend limit, no auto-reload

Free alternative: GitHub Copilot Student in the same pane.

Route 2: the vendor's extension

Part 2 of 5 · Claude Code and Codex as panels in Positron

Claude Code extension

Install and sign in

1 **Extensions:** Ctrl/Cmd + Shift + X, search Claude Code

2 **Check:** publisher Anthropic, verified badge; Install

3 **Open** your project folder and trust it

4 **Open a file, click** the spark icon or ✻ Claude Code

5 **Sign in** with Claude Pro or Max in the browser

Good to know

- Free claude.ai plan: no Claude Code
- No spark icon? Open a file, Developer: Reload Window
- Shows changes as diffs you accept

Not officially supported in Positron: if it fails, use the terminal (Part 3).

Codex extension (OpenAI)

Install and sign in

1. **Extensions:** search Codex
2. **Check:** publisher OpenAI, ID openai.chatgpt
3. **Click** Install and trust the publisher
4. **Open** the Codex panel
5. **Sign in** with your ChatGPT account

Good to know

- Works with ChatGPT Free, Go, Plus and higher; limits differ
- Same agent as the Codex CLI
- Windows: native, or WSL if needed

Not in the list? use the Codex CLI in the terminal (Part 3).

If an extension does not show up

1. **Search again** with the publisher name: Anthropic or OpenAI
2. **Switch the gallery:** setting positron.extensions.gallerySource → Open VSX, restart
3. **Install a file:** Extensions: Install from VSIX
4. **Or skip it:** the terminal route needs no extension

Only official publishers: fake look-alike extensions exist. Check the publisher and the verified badge.

Route 3: the agent in the terminal

Part 3 of 5 · install Claude Code and Codex,
sign in, work with an agent

Install Claude Code in the terminal

Windows: Positron terminal (PowerShell)

```
irm https://claude.ai/install.ps1 | iex
```

macOS: Positron terminal

```
curl -fsSL https://claude.ai/install.sh | bash
```

New terminal: check, then start

```
claude --version
claude
```

Sign in

- Pro, Max or Console account
- Code shown? Paste it

Billed to the API by mistake? if ANTHROPIC_API_KEY is set, Claude Code may use it instead of Pro. Type /status to check.

Install Codex in the terminal

Windows: Positron terminal (PowerShell)

```
powershell -ExecutionPolicy ByPass -c "irm https://chatgpt.com/codex/install.ps1 | iex"
```

macOS: Positron terminal

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

New terminal: start, check

```
codex
codex login status
```

Sign in



- Choose Sign in with ChatGPT
- Browser fails? codex login --device-auth



Works without the extension: the terminal route is the most reliable way in Positron.

Working with a coding agent

Open the project → **Start the agent** → **Ask in plain English** → **Approve changes** → **Run and check**

CLAUDE.md or AGENTS.md

```
Use polars, plotnine, great_tables
and statsmodels only. Explain each
model in two sentences. Never
read data/personal/.
```

A good request

```
Read report.qmd and list every
number in the text that is not
computed by inline code.
```

Ask-first mode: read every proposed change and command before you approve it

Small steps: one task at a time, then render and check

You own every number: agents make confident mistakes.

Route 4: MCP, the agent in your live session

Part 4 of 5 · what it does, how to switch it on,
why to be careful

What Positron's MCP server does

Run code

- in your live Python session
- labelled in the Console

See plots

- the current plot
- as an image

Use Positron

- run Positron commands
- read session history

Agents: Claude Code, Codex and Gemini CLI

Why useful: the agent sees your real data frame instead of guessing its columns

Experimental: off by default; settings and behaviour may change.

Switch it on

Claude Code

1. **Settings:** tick ai.mcp.enabled
2. **Open** the project, start Python
3. **Terminal:** claude (restart if open)
4. **Type** /mcp: positron is listed

Codex

1. **Settings:** tick ai.mcp.enabled
2. **Command Palette:** Positron MCP: Add to Coding Agent → Codex
3. **Positron terminal:** codex
4. **Check:** Positron MCP: Show Connected Agents

Then ask: “Use the positron tools to summarise the data frame choc in my Python session.”

Why to be careful with it

1. **No approval in Positron:** only the agent's own prompts stand between it and your session
2. **It sees what you loaded:** never load personal data in a session an agent can use
3. **Keep ask-first mode:** never allow every command automatically
4. **Switch it off** when you do not need it: untick ai.mcp.enabled

⚠ **For this course:** a demonstration, not a requirement. Coursework never depends on it.

Choose and use it safely

Part 5 of 5 · which route for what, costs, privacy, course rules, problems

Which route for what

I want …	Best route	Cost for a student
chat about my data, free	Posit Assistant + GitHub Copilot Student	0 USD
chat with a specific Claude or GPT model	Posit Assistant + API key	pay per use
an agent for bigger tasks, I have Claude Pro	Claude Code: terminal or extension	20 USD a month (Pro)
an agent, I have a ChatGPT account	Codex: terminal or extension	0–20 USD a month
grey code suggestions while typing	GitHub Copilot or Posit AI Pass	0 or 20 USD a month

Course default: Posit Assistant with GitHub Copilot Student. Everything else is optional.

Privacy: who sees your prompts

API key (routes 1 to 3)

- Not used for training by default
- Kept about 30 days
- Posit sees nothing

Subscription (Pro, Plus, Free)

- May be used for training while the setting is on
- Claude: claude.ai/settings/data-privacy-controls
- ChatGPT: Settings → Data Controls

Switch training off in your Claude and ChatGPT settings before you use them for coursework.

Never share, and course rules

Never in a prompt or a file

- API keys and passwords
- Names, e-mails, student or customer IDs
- Confidential or licensed data
- Exam or assessment content

Course rules

- Course default: GitHub Copilot
- Nobody has to pay for anything
- Prompt log for exercises and case study
- You verify and own every number

Prompt log: note which tool and model you used, the prompt, and what you kept or changed.

When it does not work

Problem	Fix
Claude Pro, but Positron wants a key	expected: use a Console API key, or Claude Code (routes 2, 3)
Key rejected	copy again without spaces; is there credit on the account?
No code completions	not offered by Claude or OpenAI; use GitHub Copilot
claude or codex: command not found	open a new terminal; run the installer again
Claude Code bills the API, not Pro	unset ANTHROPIC_API_KEY; check with /status
Agent cannot see the session (MCP)	ai.mcp.enabled on, Python running, restart the agent

Diagnose: Command Palette → AI: Create Diagnostic Report.

Checklist

1. **Route chosen:** Posit Assistant, extension, terminal, or MCP
2. **Account ready:** API key with a spend limit, or Claude Pro / ChatGPT account
3. **Training off** in your Claude or ChatGPT privacy settings
4. **Connected and tested** with one small question about your data
5. **Standing instructions** in CLAUDE.md or AGENTS.md
6. **Prompt log** started

Official guides

- Claude Code in VS Code and other editors
- Claude Code: install and set up
- Claude Code: data usage and privacy
- Claude plans and API prices
- Posit Assistant model providers
- OpenAI Codex (repository and install)
- Positron release notes (MCP server)

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